Heart Rate Response to Exercise

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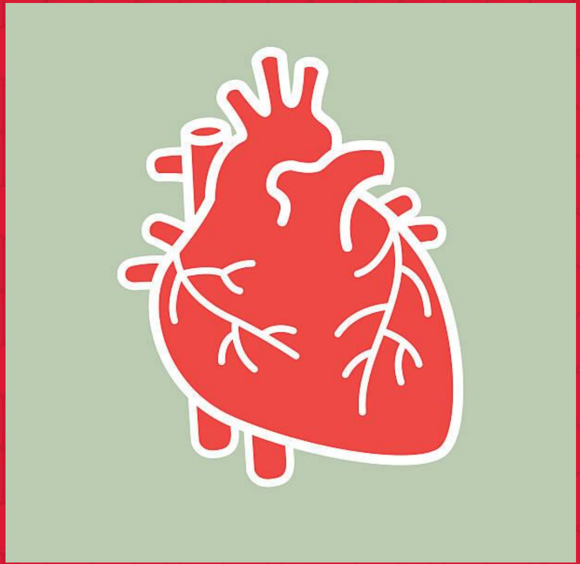

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Problem Overview / System Description

This project simulates how heart rate changes during exercise and recovery.

Heart rate increases when the body becomes active and decreases when the body rests.

This is important in biomedical engineering because it helps understand fitness and health responses.

Engineering Objective

Goal: simulate heart rate over time

Questions:

- How fast does heart rate increase?
- How quickly does it recover?

Helps understand **fitness and recovery performance**

Mathematical Model

In this project, we used basic math ideas to simulate how heart rate changes over time.

We used **exponential growth and decay patterns**:

One pattern makes the heart rate increase quickly at the beginning of exercise and then slow down as it reaches a maximum.

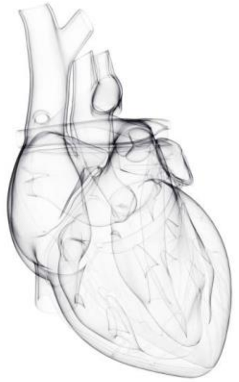
The other pattern makes the heart rate decrease quickly after exercise and then slowly return to normal.

I also used:

Variables to represent heart rate, time, and rate of change

A **rate constant (k)** to control how fast the heart rate increases and decreases

Time steps to simulate changes over minutes



Algorithm / System Logic

Start with resting heart rate

Increase time step by step

If exercising:

- heart rate **increases** over time

If exercise stops:

- heart rate **decreases** over time

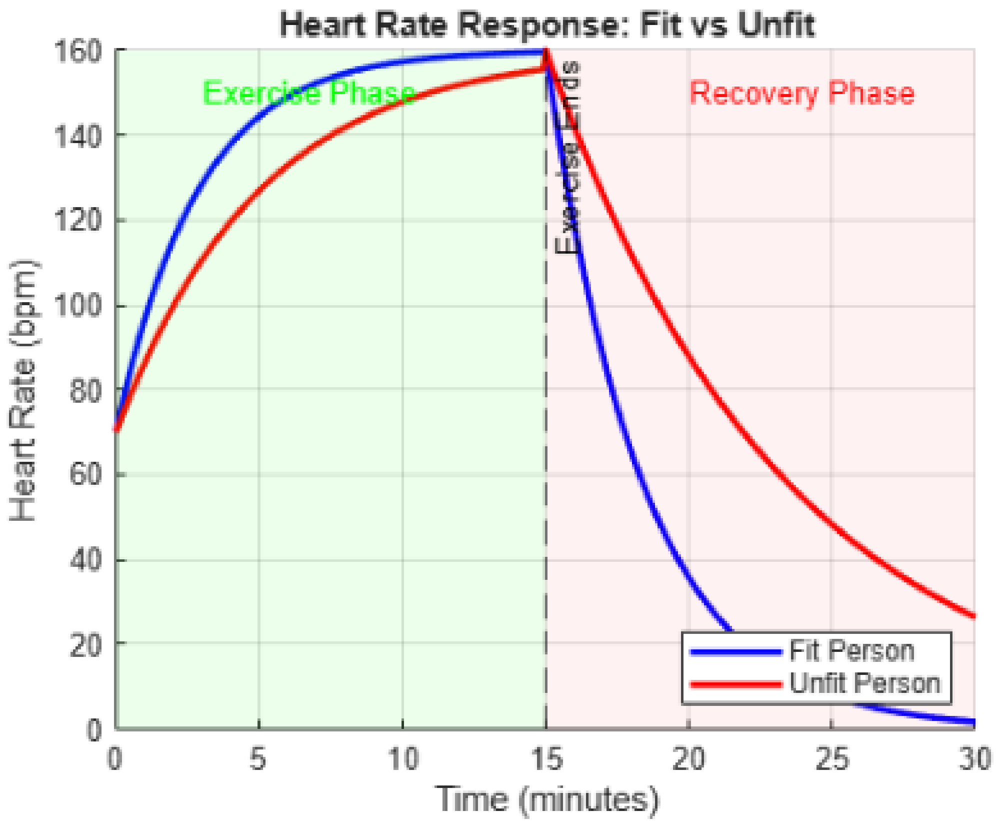
A **rate constant** controls how fast changes happen

MATLAB Implementation

- Defined time values in minutes
- Set resting and maximum heart rate
- Used formulas to simulate changes over time
- Used a loop to calculate values step-by-step
- Plotted heart rate vs time

Simulation Results (Plots)

- Heart rate **increases** during exercise
- Reaches a peak value
- Then **decreases** during recovery
- The graph shows a smooth rise and fall over time



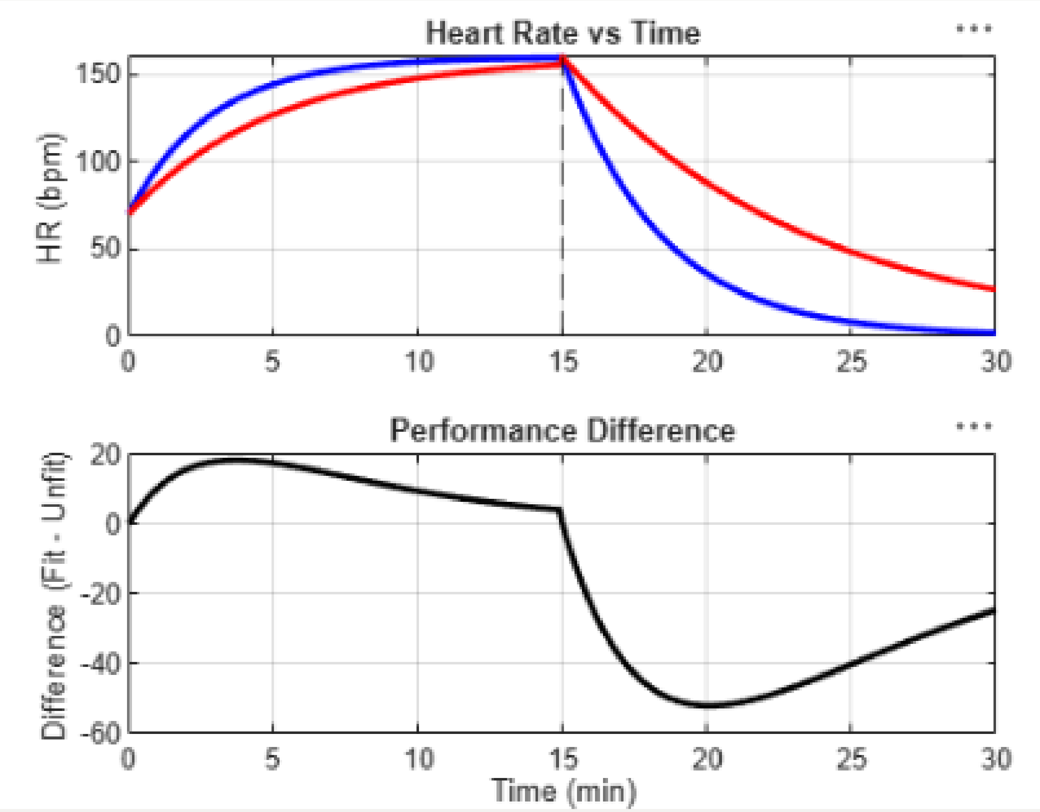
Analysis of Results

Fit vs Unfit comparison

- Fit person:
  - faster recovery
  - smoother heart rate curve
- Unfit person:
  - slower recovery
  - higher sustained heart rate

<u>Engineering insight</u>

- Higher k = better cardiovascular efficiency
- Lower k = slower physiological response



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Parameter/(k) Explanation

- The rate constant controls how fast changes happen
- **Higher value = faster increase** and faster recovery
- **Lower value = slower changes**
- It represents fitness level or response speed

Model Limitations

This is a **simplified** model

It does not include:

- age differences
- daily habits
- medical conditions

Real human data is more complex

Engineering Relevance

This type of model is used in:

- fitness trackers
- health monitoring devices
- biomedical research

It helps engineers understand how the human body responds to exercise.

Conclusion

We created a model of heart rate during exercise.

We learned how mathematical equations represent real systems.

Key takeaway:

Engineering models help us understand and predict cardiovascular behavior.